

Haokai Ding

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Underactuated mechanism design for grasping and aerial manipulation. 9 papers in IROS, CASE, ROBIO, ICCMA, and other venues.

EDUCATION

Mohamed bin Zayed University of Artificial Intelligence (MBZUAI) *Fall 2026 – (Incoming)*
M.Sc. in Robotics

Tsinghua University – Shenzhen X-Institute *Sept 2023 – Present*
Tsien Excellence Engineering Program (TEEP), Joint Training (Underactuated Robotics)

Shenzhen Technology University (SZTU) *Sept 2022 – Jul 2026*
B.Eng. in Electronic Science & Technology; GPA: 84.6/100
Research focus: LIBS spectroscopy; robotics systems.

PUBLICATIONS

Legend: [†] Co-first author; ^{*} Corresponding author. Listed in reverse chronological order by IEEE Xplore date added.

Selected Conference Papers (Robotics)

- [1] An Idle-Stroke Underactuated Gripper with Independent Double-Parallelogram Linkages for Pinching and Adaptive Grasping
Y. Chen, H. Luan, **H. Ding**, W. Zhang^{*}
IEEE RAAI 2025, Singapore Mar 2026
- [2] BioCrest – A Flexible Adaptive Robotic Hand Based on Idle-Stroke Mechanism
H. Ding[†], X. Zhong, W. Huang, T. Yang^{*}, W. Zhang^{*}
IEEE ROBIO 2025, Bangkok, Thailand Feb 2026
- [3] GLINT: An Idle-Stroke Grasp-and-Lift Hand for In-Hand Manipulation
H. Ding, H. Luan, T. Yang^{*}, W. Zhang^{*}
IEEE ICCMA 2025, Macau, China Feb 2026
- [4] Peaucellier Gripper: A Novel Underactuated Gripper for Linear Pinching and Self-Adaptive Grasp
H. Ding[†], J. Fan, Z. Zhu, Y. Zhao, K. Chen^{*}, W. Zhang^{*}
IEEE ICARM 2025, Hangzhou, China Dec 2025
- [5] An Underactuated Gripper with Grasshopper-Inspired Linkages and Delay Triggering for Pinching and Adaptive Grasping
H. Ding[†], Y. Chen[†], D. Jia, W. Zhang^{*}
IEEE ICIA 2025, Wuhan, China Dec 2025
- [6] A Novel Gripper with Semi-Peaucellier Linkage and Idle-Stroke Mechanism for Linear Pinching and Self-Adaptive Grasping
H. Ding[†], W. Zhang^{*}
IEEE/RSJ IROS 2025, Hangzhou, China Nov 2025
- [7] Semi-Peaucellier Linkage and Differential Mechanism for Linear Pinching and Self-Adaptive Grasping
H. Ding[†], Z. Chen, T. Yang^{*}, W. Zhang^{*}
IEEE CASE 2025, Los Angeles, USA Sep 2025

Other Publications

- [8] Recent Progress on the Research of 3D Printing in Aqueous Zinc-Ion Batteries
Y. Liu[†], **H. Ding**[†], H. Chen, H. Gao, J. Yu, F. Mo, N. Wang^{*}
Polymers (MDPI) 17(15):2136. [SCI, JCR Q1] Aug 2025

[9] Rapid Analysis of Heavy Metal Element Adsorption by SCG Based on LIBS Technology
H. Xie, L. You, X. Fang, L. Li, **H. Ding**, Z. Zhou, G. Zhang, D. Zhang
E3S Web of Conf., 615 (2025) 02010 Feb 2025

RESEARCH & PROJECTS

Underactuated Robotic Gripper Research
Shenzhen X-Institute, Tsinghua University

2023 – Present

Designed and tested underactuated robotic grippers with semi-Peaucellier linkage. Focus: idle-stroke / delay-triggering mechanisms; validated prototypes via simulation and experiments.

LIBS Spectroscopy for Heavy Metal Analysis
Shenzhen Technology University

2022 – 2023

Built and optimized LIBS system for rapid adsorption analysis. Focus: improved calibration pipeline and enhanced acquisition accuracy.

EXPERIENCE

Visiting Student – State Key Laboratory of Mechanical System and Vibration, Shanghai Jiao Tong University
Advisor: Prof. Wei Dong. Research focus: UAV perching and aerial manipulation.

Feb 2026 – Present

HONORS & AWARDS

National Scholarship, Ministry of Education of China	2025
Undergraduate Champion, ASME Student Mechanism & Robot Design Competition	2025
Conference Session Chair, IEEE / RSJ IROS 2025	2025
President’s Award, Shenzhen Technology University	2024
Research and Innovation Award (First Prize), SZTU	2024
Gold Prize, 9th China “Internet+” Innovation Competition (Guangdong)	2023
Second Prize, 17th “Challenge Cup” (Guangdong)	2023
Research and Innovation Award (First Prize), SZTU	2023
Chen Hsong Scholarship, SZTU	2023

SERVICE

- Conference reviewer for IEEE / RSJ IROS and IEEE CASE.
- Conference session chair for IEEE / RSJ IROS 2025.

SKILLS

Programming: C/C++, Python, MATLAB
Tools: SolidWorks, Arduino, Linux, L^AT_EX
Languages: Chinese (native), English (fluent)